

Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title:** Book Store
- **Team Members :** Bramhanapalli Mohitha (Full stack developer), Chaitanya Potla, Adharsh Ravula, Bokkisar Priyankisho Reddy, Govardhan Ugandhar

2. Project Overview

- **Purpose:**

To design and build an intuitive, end-to-end e-commerce platform for book lovers, enabling seamless book discovery, shopping cart management, and administrative control over inventory.

- **Features:**

- **Dynamic Catalog:** Browse, search, and filter books by genre, author, price, and ratings.
- **Shopping Cart & Checkout:** Add/remove items, update quantities, and simulate secure checkouts.
- **Admin Panel:** Complete CRUD operations for Administrators to add, update, or remove books and view total sales.
- **User Reviews:** Allows registered users to leave ratings and text reviews on individual book profiles.

3. Architecture

- **Frontend:** Built with React.js using functional components and hooks for state handling, ensuring a responsive single-page experience.
- **Backend:** A RESTful API built on Node.js and Express.js handling store operations, order collection, and user routing.
- **Database:** MongoDB utilizes flexible document models to link store inventories, transaction logs, and user credentials.

4. Setup Instructions

• Prerequisites:

Ensure your system meets these specifications before launching the application:

Hardware Requirements

- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or better.
- **RAM:** Minimum 8 GB (16 GB recommended).
- **Storage:** At least 1 GB free space.
- **Display:** 1366x768 or higher for layout visualization.

Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Node.js:** v16 or above (includes npm v8 or above).
- **MongoDB:** Community Edition installed locally or a MongoDB Atlas cloud instance.
- **Git & GitHub:** Installed for code version management.

• Installation:

1.Clone the Repository:

Backend Setup:

- Open a terminal, navigate to the server folder: `cd server`
- Install dependencies: `npm install`
- Create a `.env` file in the server directory and add your variables: ○
Code snippet

`PORT=5000`

`MONGO_URI=mongodb://localhost:27017/bookstoreDB`

`JWT_SECRET=your_store_secret_key`

Frontend Setup:

- Open a new terminal, navigate to the client folder: `cd client`
- Install packages: `npm install`
- Create a `.env` file in the client directory:
- Code snippet

`REACT_APP_API_URL=http://localhost:5000/api`

5.Folder Structure

Plaintext bookverse-store/

```
|— client/ (React Frontend) [cite: 18]
|   |— public/
|   |— src/
|       |— components/ # BookCard, Navbar, CartItem, ProtectedRoute
[cite: 18]
|       |— pages/      # Home, BookDetails, Cart, AdminDashboard [cite:
18]
|       |— context/    # CartContext, AuthContext [cite: 18]
|       |— App.js
|— server/ (Node/Express Backend) [cite: 19]
|   |— config/         # Database connection file [cite: 19]
|   |— controllers/    # bookController, orderController, authController
[cite: 19]
|   |— models/         # Book.js, User.js, Order.js (Mongoose Schemas)
[cite: 19]
|   |— routes/         # API endpoint routes [cite: 19]
|   |— server.js
|— README.md
```

6.Running the Application

To execute the application locally, run these commands in separate terminals:

- Start Backend: Run `npm start` in the server directory. (*Runs on `http://localhost:5000`*)
- Start Frontend: Run `npm start` in the client directory. (*Runs on `http://localhost:3000`*)

7. API Documentation

The backend exposes the following primary endpoints for UI data consumption:

Meth od	Endpoint	Description	Request Body / Params	Example Response
GET	/api/books	Fetches all books from the store	<i>None</i>	[{ _id, title, price... }]
GET	/api/books/:id	Fetches details of a specific book	id (Param)	{ title, author, description }
POST	/api/books	Adds a new book (<i>Admin Only</i>)	{ title, author, price, stock }	{ message: "Book Added" }

8. Authentication

- Mechanism: Managed via JSON Web Tokens (JWT).
- Workflow: Upon registration or login, the user's credentials are verified. The server signs a unique JWT that is sent back and stored in the application's memory/localStorage.
- Authorization: Role-based checks distinguish regular buyers from store Admins. Protected administrative routes require a token with verified administrative clearance to mutate catalog data.

9. User Interface

10. Testing

- **Strategy:** Endpoint confirmation and route security analysis.
- **Tools:** Postman was utilized during production to simulate payloads, trace response headers, and assure middleware correctly flags missing tokens on protected actions.

11. Screenshots or Demo

BooksWorld[Home](#)[Login](#)[Sign Up](#)

Join BooksWorld

Create your account to get started

Register as a...

User

Seller

Admin

Full Name

Email address

Password

Create Account

Already have an account? [Sign in here](#)

BooksWorld[Home](#)[Login](#)[Sign Up](#)

Welcome Back to BooksWorld

Sign in to your account to continue

Login as a...

User

Seller

Admin

Email address

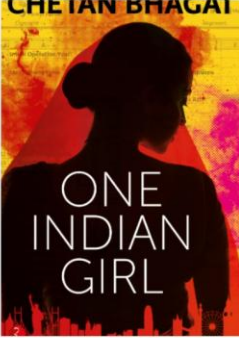
sell@gmail.com

Password

Sign In

Don't have an account? [Sign up here](#)

BooksWorld[Home](#)[Login](#)[Sign Up](#)



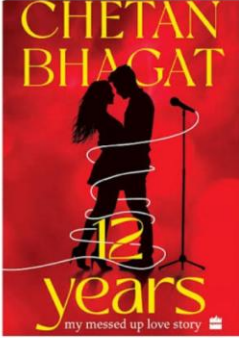
NON-FICTION

ONE INDIAN GIRL

Chethan Bhagath

₹80

[Add to Cart](#)



FICTION

12 YEARS

Chethan Bhagath

₹300

[Add to Cart](#)



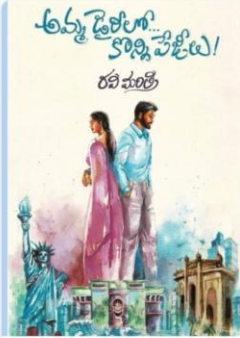
FICTION

Dheera Sameere Ganga...

Ravi Mantri

₹220

[Add to Cart](#)



FICTION

Amma Diarylo Konni...

Ravi Mantri

₹118

[Add to Cart](#)

BooksWorld

[Home](#)[My Inventory](#)[Add Book](#)[Orders](#)

Logout

My Inventory

NON-FICTION
ONE INDIAN GIRL
₹80
150 in stock
[Delete](#)

FICTION
Amma Diarylo Konni...
₹118
200 in stock
[Delete](#)

BooksWorld

[Home](#)[Cart](#)[My Orders](#)

Logout

My Orders

Order ID: ORD-615553

₹108

(Qty: 1)

BooksWorld

[Home](#)[Overview](#)[All Books](#)[Sellers](#)[Users](#)

Logout

Welcome back, Admin

Platform Overview Dashboard

Total Books

4

Active Sellers

2

Registered Users

3

Platform Distribution

Activity Overview

BooksWorld

[Home](#) [My Inventory](#) [Add Book](#) [Orders](#) [Logout](#)

List a New Book

Book Title

Author Name

Genre

Fiction

Price (₹)

Stock Quantity

Description

Cover Image

Choose File

No file chosen

Publish Book to Store

BooksWorld

[Home](#) [Overview](#) [All Books](#) [Sellers](#) [Users](#) [Logout](#)

Manage Catalog

TITLE	AUTHOR	PRICE	ACTIONS
ONE INDIAN GIRL	Chethan Bhagath	₹80	Remove
12 YEARS	Chethan Bhagath	₹300	Remove
Dheera Sameere Ganga Teere	Ravi Mantri	₹220	Remove
Amma Diarylo Konni Pageelu	Ravi Mantri	₹118	Remove

BooksWorld

[Home](#) [Overview](#) [All Books](#) [Sellers](#) [Users](#) [Logout](#)

Manage Users

NAME	EMAIL
user	user@gmail.com
mohi	mohi@gmail.com
Adminuu	adminuu@gmail.com

BooksWorld			
		Home	Overview
		All Books	Sellers
		Users	Logout
Manage Sellers			
NAME	EMAIL	STATUS	ACTIONS
Seller	seller@gmail.com	Approved	
sell	sell@gmail.com	Pending	Approve

12. Known Issues

- **Cart Persistence:** In rare instances, logging out clear-wipes items inside an active guest cart instead of persisting it temporarily.
- **Search Latency:** Large text-queries against the catalog experience a minor delay due to a lack of text indexing configurations on MongoDB Atlas.

13. Future Enhancements

- **Stripe Payment Gateway:** Swapping out simulated orders for concrete, real-world card payments.
- **Audiobook/PDF Previewer:** Integrating an interactive widget allowing shoppers to read a 5-page snippet or sample audio clips before making purchases.
- **Recommendation Engine:** A customized "You Might Also Like..." algorithm analyzing purchase patterns and favorite genres